

Data Audit

No feature enters scope until its data is named and its logic verified.

Rule: every claim below is traced to a specific source. If the data isn't there, the feature is cut or degraded — not promised and hoped for.
Status key:  data confirmed, claim safe ·  data exists but constrains the claim ·  data missing, needs fallback

Q1 · F1 — "Where does your water come from?"

Needs	Mapping from a location → supply zone → reservoir → treatment plant → supply type (24x7 or intermittent)
Exists	City-level only: sources are Pench/Kanhan/Gorewada; OCW operates WTPs at Gorewada and Kanhan; 68 reservoirs, 2000+ km network, 3M+ consumers. The area→reservoir mapping is not published anywhere we can find.
Can claim	Which broad source system feeds the city; the network's structure
Cannot claim	"Your specific reservoir is X" — we do not have it
Fallback	Degrade to NMC's administrative zones. Say " <i>your zone</i> ", labelled approximate. Hard cutoff: if the reservoir command areas aren't in hand by hour 2 of the build, ship zones and move on. Do not spend hackathon hours hunting a PDF.
Status	 →  with fallback

Q1 · F2 — Usability verdict

Needs	Measured parameters at a location, judged against a standard
Exists	CGWB district sampling: 263 shallow + 84 deep points, with nitrate, fluoride, EC, pH, hardness, chloride . MPCB: BOD/DO at 5 river stations . OCW: sample fitness, city-wide aggregate only .
Critical gap	No bacteriological data anywhere in our sources. IS 10500 requires absence of E. coli and total coliforms for potability.
Can claim	"Exceeds the IS 10500 limit for [named parameter], measured [date], [distance] from your location"
Cannot claim	" Safe to drink. " Ever. Not at any point, not with any caveat.
The verdict has three states, and none of them is "safe"	① Exceeds limit for [parameter] · ② No exceedance among the [n] parameters tested here — not tested for bacteriological quality · ③ Not tested
Why this is better	It removes the liability exposure, it matches our own integrity slide, and refusing to say "safe" when we can't is the credibility move the whole project rests on
Status	 — solid, but only with the asymmetric verdict

Q1 · F3 — Crop verdict (*strongest feature*)

Needs	Water EC and pH + published crop salt-tolerance relationships
Exists	EC and pH from CGWB points. FAO-29 publishes, per crop, a threshold EC and a percent yield-loss slope above it. These are standard tables used by ICAR and state agriculture departments.
Technical caveat a professor may probe	FAO tables are given in both ECe (soil saturation extract) and ECw (irrigation water). Use the ECw column and state the assumed leaching fraction. Don't silently mix the two — that's the one place this feature can be attacked.
Standards split	CPCB Class E (EC < 2250 µS/cm) for the statutory pass/fail verdict; FAO-29 for the crop-specific yield estimate. Two standards, two different jobs — say which is doing which.
Can claim	"At this measured EC, FAO-29 puts [crop] at [x]% yield reduction" — computed from the published table
Cannot claim	Yield predictions for a specific farm. Salinity is one of many factors
⚠ Read the actual FAO-29 table for the threshold and slope of each crop. Do not use any percentage I quoted in earlier documents — I did not verify those against the source. The qualitative contrast (citrus sensitive, cotton tolerant) is large and robust; the exact numbers must come from the table	
Status	🌀

Q1 · F4 — Map of what is actually known

Exists	~347 CGWB points, 5 MPCB stations, OCW aggregate, plus anything we add
Can claim	Every real measurement, shown as a point, with its date and distance from the user
Must not do	Draw a smooth colour surface. With this density, interpolation invents knowledge — and it's the exact sin we accuse OCW of
The trap	Our own answer for most residents is "grey / we don't know." A judge will ask what we actually <i>add</i> over OCW's single number
The answer	Don't pitch the map — pitch the instrument. On stage, enter a reading and watch a grey cell resolve into a verdict. The map is the <i>before</i> . The demo has to show the <i>after</i> . And reframe the line: " the grey isn't the finding, it's the worklist — and here's what it costs to fill one cell. "
Status	🌀 with the demo fix

Q1 · F5 — Which outfalls are upstream of you

Needs	Outfall locations + river topology + monitoring values
Exists — confirmed today	All 12 outfalls have identifiable place names and can be geocoded by hand:
Nag (9)	Dande Hospital/Ravi Nagar Chowk · Bore Nalla behind Naivadyam · Untkhana Bridge · Jagnade Chowk (Nandanvan) · Super Store, Jagnade Chowk · St. Xavier School (Vyankatesh Nagar) · Hasanbagh nr Vyankatesh Nagar · Hudkeshwar Nalla nr bridge · Gandhi Nagar behind LAD College
Pili (3)	Chambhar Nalla, Sharda Ispat-Kalamna bridge (Indora) · Nagpur-Koradi railway crossing · Zingabai Takli nr St. Vincent Pallati School
Stations (5)	Ambazari · Bhandewadi Bridge · Asoli Bridge (Bhandara Rd) — Nag · Wanjra Layout (Kamptee Rd) · Mankapur (Koradi Rd) — Pili
Hard constraint found today	No discharge volume is published for any individual outfall. And with only 3 stations on the Nag, multiple outfalls sit between any two stations
Can claim	"These 3 outfalls lie upstream of your point", and "BOD rises from 4-13 at Ambazari to an average of 57 at Bhandewadi — these are the outfalls in that segment"
Cannot claim	Attribution to any single outfall. No "outfall 3 contributes 40%." That number does not exist and I should not have used it as an illustration earlier
Correct framing	Segment-level attribution, not outfall-level
Status	🌐 — real, but bounded

Q1 · F6 — Citizen sample entry

The error	A TDS meter is a conductivity probe. It cannot measure nitrate, fluoride, coliforms or metals — i.e. it cannot measure any contaminant we lead with. It measures salinity, which our own data shows is the <i>less</i> common problem (5.3% of shallow samples)
What it legitimately does	① Irrigation screening — FAO irrigation limits are largely EC-based, so the crop verdict genuinely runs on a TDS meter. ② Change detection — " <i>your borewell's TDS jumped 40% since last reading, get it lab-tested.</i> " A trend from a cheap instrument is real information
Optional second tier	Nitrate and fluoride field strips run roughly ₹5-15 per test
Rule	Citizen readings never produce a potability verdict. They produce an irrigation verdict, a trend, or a "get this tested" flag
Must appear on the entry screen	What this instrument can and cannot detect. Same honesty move as the grey map, applied where it costs us something
Status	🌐 — valid once rescoped

Q2 · Rainfall response

Needs	Per-location flood behaviour
Exists	30 m DEM · OSM: 89 drains, 135 streams, 54 rivers, 24 canals (only 9 have a width tag) · reported locations from r/nagpur, Google, personal knowledge · Open-Meteo rainfall
The contradiction we nearly shipped	" <i>This junction floods above 40 mm</i> " is precisely the labelled dataset we said doesn't exist. The number would be invented or self-generated — both of which we declared indefensible in the same document
Can claim	An ordinal rank. " <i>This junction is in the top 5% of the city for ponding risk, given catchment area, drain density and sink depth.</i> " That is exactly the "screening, not design discharge" language we already committed to, and the DEM + OSM network supports it
Cannot claim	Any numeric rainfall threshold. Any depth
Second contradiction	We rejected safe-route navigation because it invites the Google comparison — then led Q2 with the Google comparison. Never put "route" next to "Maps." Our defensible claim is timing: <i>we warn before, not during</i>
Status	🌐 — ordinal only

Q3 · Is anyone fixing it

Exists at start	Nothing. Entirely crowdsourced
The problem	It's our most original idea, it's scheduled last, and a 30-day loop cannot be demonstrated in a 3-minute pitch — highest differentiation, lowest chance of existing, zero demo surface
Decide before the 17th — pick one	(a) Don't build it. Give it one spoken sentence: " <i>and the version-two hook is the question nobody asks — thirty days later, did it work?</i> " Eight seconds, buys the originality credit, zero build risk. (b) Build only the follow-up mechanic against seeded data with shifted timestamps, demo as a time-travel view
Do not	Leave it as "then Q3." That's how it becomes a screenshot of our best idea, which reads as vapour
Status	🌀 until decided

Claims withdrawn from earlier documents

Withdrawn	Reason
"Deep aquifer is worse — the opposite of what everyone assumes"	13.3% vs 20.2%, unpaired, n=263 vs 84 → z ≈ 1.55, p ≈ 0.12 . CIs overlap heavily. Survives only as " <i>in this dataset deep samples exceeded more often — unpaired, so a question worth asking, not a conclusion</i> "
"345 – 100 = 265 MLD untreated"	Reconcile it properly: 345 generated · 100 MLD capacity running at 80% → ~80 treated → ~265 untreated . Anyone can do the subtraction otherwise. Also 2011 data — assign a named person to verify current capacity on 16 Aug or drop it from the spoken pitch
"Outfall 3 contributes 40% of BOD load"	No per-outfall volumes exist. Fabricated illustration. Deleted
"~30% orange yield loss at EC 3.1"	Not verified against the FAO-29 table. Recompute from source
"Nobody measures where you are"	Unpublished ≠ non-existent — and the person who can refute it may be on the panel. Reframe to " no resident can reach it " and pre-load the refutation as an ask: " <i>if that data exists, we'd take the feed tomorrow</i> "
"Two hours of compute for any city"	Never benchmarked

Still missing — one slide, not a feature

Architecture and the ask. Nothing anywhere answers "*what was technically hard about this build?*" — which is the question the professors are actually scoring. Have one sentence each on the three non-trivial pieces: the **upstream-outfall spatial join along the river network**, the **multi-standard verdict engine** reconciling IS 10500 / CPCB / FAO, and **DEM-based sink delineation** for Q2.

And for the municipal half: who owns it after, what it costs to run, how citizen submissions are moderated, and **what we are asking NMC for** — a zone-level feed from OCW, and read access to works data. **End the pitch on the ask.** Officials respond to being asked for something concrete far better than to being shown a map.